

PGI-RC Correction: Results

Corrected inequality of opportunity estimates

In MoBa, Howe’s correction is the preferred one, because it is based on the same data and includes both between and within family SNP heritability estimates. In WLS, the choice is less clear, because Becker’s correction is based on WLS data, but within family SNP heritability is inferred from Howe et al. We could keep Howe’s correction in WLS for internal consistency of between and within SNP heritability.

The correction substantially reduces IOLib — because the noisy PGI currently over-attributes variance to shared family environment — and modestly increases IORad through within-family variance. The gap roughly doubles when corrections are applied. The effect size of the correction is consistent with Becker et al. (2021), see Table 3 in their paper for an example of correction.

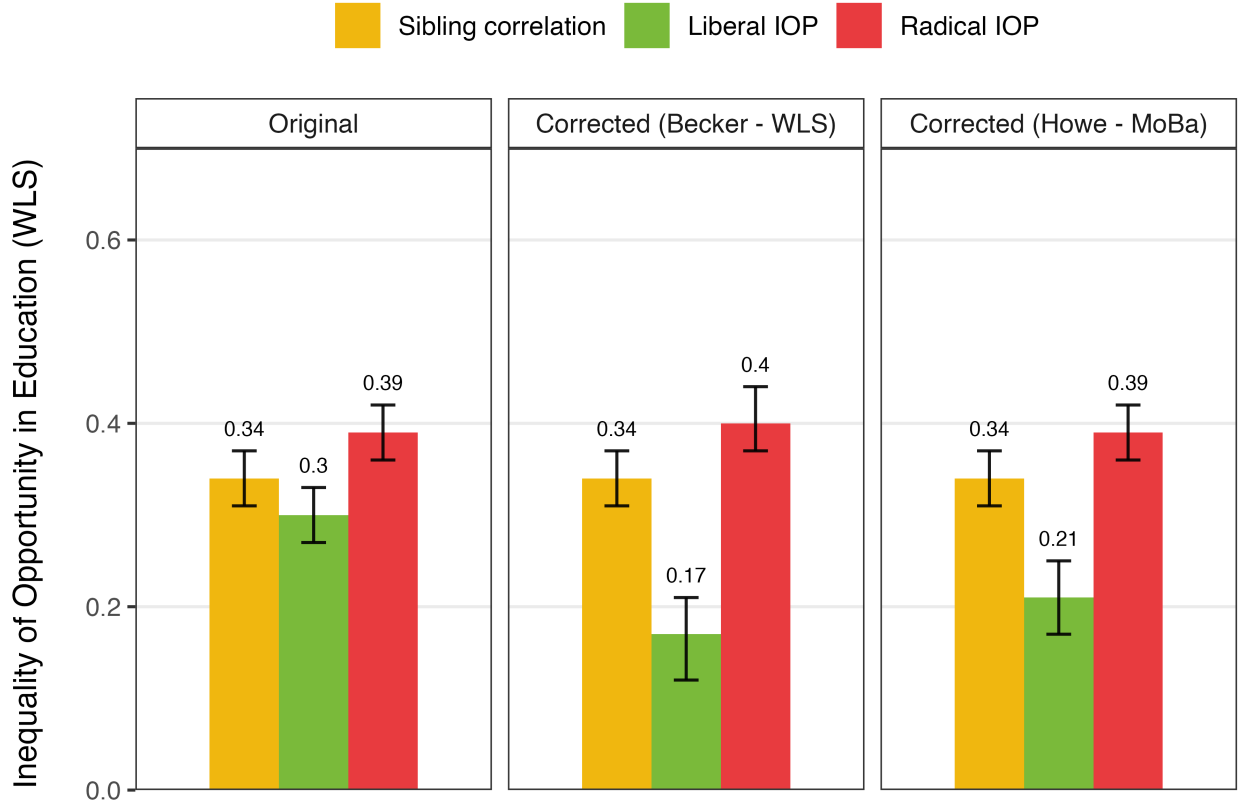


Figure 1: WLS: corrected indices

	h^2_{between}	h^2_{within}	ρ_{between}	ρ_{within}	Sibcorr	IOLib	IORad	Gap
Original	—	—	—	—	0.34	0.30	0.39	0.09
Corrected (Becker – WLS)	0.119	0.053	1.939	1.255	0.34	0.17	0.40	0.23
Corrected (Howe – MoBa)	0.090	0.040	1.686	1.091	0.34	0.21	0.39	0.18

WLS data parameters: $R^2_{\text{between}} = 0.094$, $R^2_{\text{within}} = 0.051$, $ICC = 0.337$, $R^2_{\text{pop}} = 0.065$.

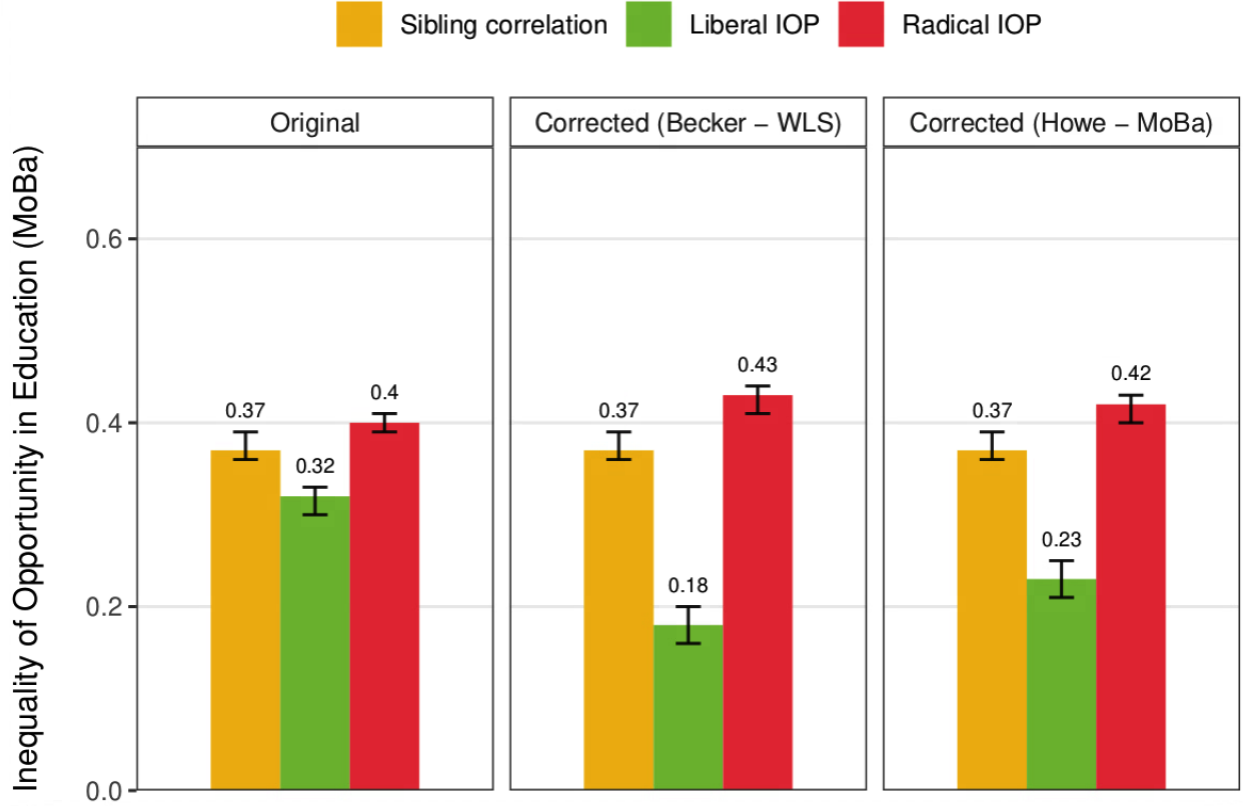


Figure 2: MoBa: corrected indices

	h^2_{between}	h^2_{within}	ρ_{between}	ρ_{within}	Sibcorr	IOLib	IORad	Gap
Original	—	—	—	—	0.37	0.32	0.40	0.08
Corrected (Becker – WLS)	0.119	0.053	1.752	1.975	0.37	0.18	0.43	0.25
Corrected (Howe – MoBa)	0.090	0.040	1.523	1.717	0.37	0.23	0.42	0.19

MoBa data parameters: $R^2_{\text{between}} = 0.104$, $R^2_{\text{within}} = 0.022$, $ICC = 0.374$, $R^2_{\text{pop}} = 0.052$.

Check: single- ρ vs. decomposed correction

The decomposed correction (ρ_{between} , ρ_{within} separately) used to compute the corrected version of each index should be consistent with a check where a single ρ_{pop} scales the radical-liberal gap, accounting for the total genetic contribution (between and within families):

$$\text{IORad}_{\text{RC}} - \text{IOLib}_{\text{RC}} = \rho_{\text{between}}^2 \Delta_{\text{between}} + \rho_{\text{within}}^2 \Delta_{\text{within}} = \rho_{\text{pop}}^2 \Delta_{\text{observed}}$$

Single ρ_{pop} is either:

- implied from the data, using h^2_{pop} values reported in Becker et al (2021) and in Howe et al (2022)
- taken as directly reported in Becker et al (2021)

The gap check validates the decomposed correction. In **WLS** the decomposed Becker result (0.24) matches Becker's directly reported $\rho = 1.649$ exactly, and is close to the implied single- ρ value (0.23). The Howe

decomposed result (0.18) is also close to the Howe single- ρ check (0.17). This internal consistency confirms that the between/within decomposition of ρ is coherent with the population-level estimate.

In **MoBa** the match is looser for Becker: the decomposed result (0.24) sits between the reported single- ρ (0.21) and the implied single- ρ from our R^2_{pop} (0.26). The Howe decomposed result (0.18) matches the Howe single- ρ (0.19) closely. The small discrepancy for Becker in MoBa reflects the fact that Becker's h^2 estimate is WLS-specific and the R^2 is estimated in MoBa, introducing a cross-study mismatch for that scenario.

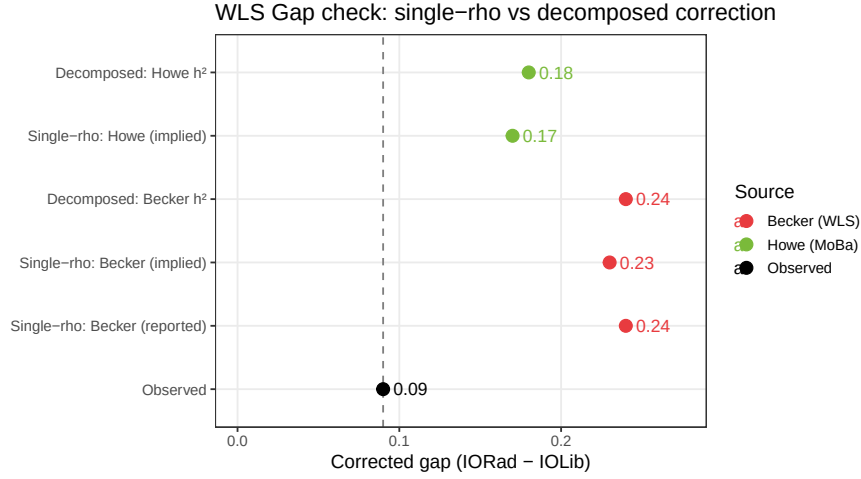


Figure 3: WLS: decomposition check

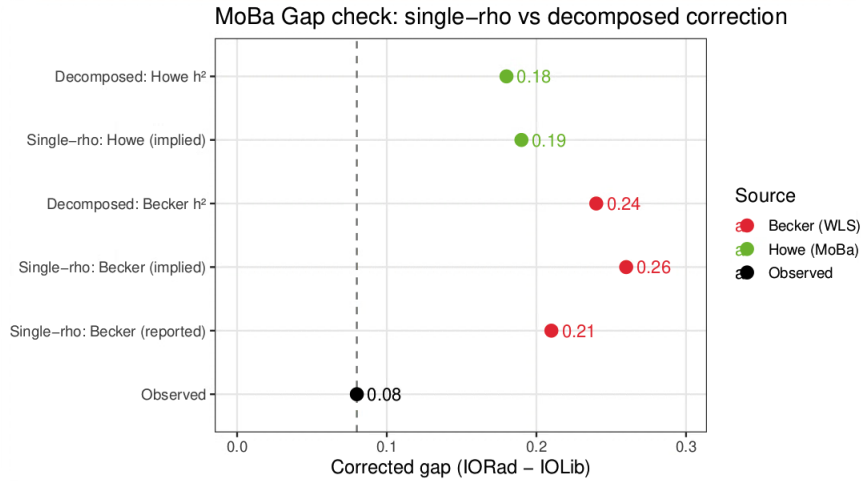


Figure 4: MoBa: decomposition check